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17 SEPTEMBER 1958
SUPERSEDING
MIL-C-10392A
13 JANUARY 1953

MILITARY SPECIFICATION

CABLES, SPECIAL PURPOSE, ELECTRICAL (MINIATURE)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers miniature, special purpose, electrical cables used at potentials up to 300 volts (rms) a.c. (see 3.2 for individual types).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

J-C-98 — Cable and Wire, Insulated; Methods of Sampling and Testing.

QQ-S-781 — Strapping, Flat; Steel.

LLL-F-291 — Fiberboard; Corrugated, Single-Face (Flexible).

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

PPP-T-97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood-Cleated Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-B-640 — Boxes, Folding, Fiberboard, Corrugated, Triple Wall.

MILITARY

MIL-P-116 — Preservation, Methods of.

FSC 6145

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MIL–W–3861 — Wire, Electrical (Bare Copper).

MIL–I–3930 — Insulating and Jacketing Compounds, Electrical (For Cable, Cord, and Wire).

MIL–C–13777 — Cable, Special Purpose, Electrical.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL MOTOR FREIGHT CLASSIFICATION No. A–3 AND No. 14

(Application for copies should be addressed to the American Trucking Association, Traffic Department, 1424 16th Street, N. W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material. The material used in the cables shall be as hereinafter specified.

3.2 General. The cables shall be constructed as follows:

Conductors

Cable type	Total No.	Size No. each	AWG size (nominal)	Circular mils (nominal)	Cable diameter (inch nominal)
WD–27A/U	2	2	24	404	0.146
WT–15A/U	3	3	24	404	0.212
WF–11A/U	4	4	24	404	0.212
WM–59A/U	5	5	24	404	0.212
WM–60A/U	6	6	24	404	0.252
WM–69B/U	6	2	18	1620	0.262
		4	24	404	
WM–111A/U	7	2	18	1620	0.280
		5	24	404	
WM–61A/U	7	7	24	404	0.252
WM–62A/U	8	8	24	404	0.262
WM–70B/U	8	2	18	1620	0.285
			24	404	
WM–63A/U	9	9	24	404	0.280
WM–64A/U	10	10	24	404	0.297

3.3 Conductor. The conductor shall be constructed as follows.

3.3.1 *Conductor construction, 404 circular-mils.* Conductors having a nominal cross-sectional area of 404 circular-mils (No. 24 AWG) shall consist of 41 strands, bunch stranded, with a maximum length of lay of 0.6-inch.

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3.3.2 *Conductor construction, 1620 circular mils.* Conductors having a nominal cross-sectional area of 1,620 circular-mils (No. 18 AWG) shall consist of 168 strands, rope lay, consisting of seven bunch-stranded members. Each bunch-stranded member shall contain 24 strands, bunch-stranded with a maximum length of lay of 0.6 inch. Six of these members shall be twisted concentrically around one member, with a maximum length of lay of 0.8 inch to form the rope-lay conductor.

3.4 Conductor strands. The strands comprising the stranded conductors shall have a cross-sectional area of 9.9 circular-mils (No. 40 AWG). The diameter of the strands shall be 0.0031 inches plus 0.0003 inch, minus 0.0001 inch. The strands shall be tinned, annealed, cadmium-copper alloy having a 1-percent nominal content of cadmium.

3.4.1 *Coating.* Each strand shall be coated with tin as specified in MIL–W–3861 for No. 40 AWG wire.

3.4.1.1 *Continuity of coating.* The coated strands shall show no exposed cadmium-copper when tested as specified in 4.7.

3.4.1.2 *Adhesion of coating.* The coating shall adhere firmly to the surface of the cadmium-copper strand (see 4.8).

3.4.2 *Conductor strand tensile strength and elongation.* The tensile strength and elongation of the coated strands prior to stranding of conductor shall be as follows:

Tensile strength, maximum	*Elongation in 10 inches*
50,000 psi	13 percent min.

3.4.3 *Conductor strand resistance.* Each conductor strand shall have a d.c. resistance per 1,000 feet of 1,310 ohms, maximum, at a temperature at or corrected to 20° C. If the resistance value is lower or equal to that specified, when the measurement is made at a temperature greater than 20° C. no correction factor need be employed.

3.5 Inner separator. An inner separator shall be applied over each conductor. The separator shall consist of a suitable number of ends of acetate rayon, saponified acetate rayon, or nylon yarn closely wound to cover the conductor. If however, the insulating compound applied in accordance with 3.7 is free stripping, the separator may be omitted.

3.6 Insulation. A Buna S insulating compound, Type IS, conforming to Specification MIL–I–3930 shall be applied over the conductor or inner separator, when present, to a minimum thickness of 0.010 inch. Insulated conductors having a nominal area of 404 circular mils (No. 24 AWG) shall have a maximum diameter of 0.059 inch. Insulated conductors having a nominal area of 1,620 circular mils (No. 18 AWG) shall have a maximum diameter of 0.097 inch.

3.7 Insulating compound. Each insulated conductor shall be covered with a differently colored insulating compound, as follows.

3.7.1 *Color coding of cables having all 404 circular mil conductors.* Cables having all 404 circular mil conductors shall use the following colors, in the order of preference as listed below:

Order	*Color*
1	White
2	Black
3	Red
4	Green
5	Orange
6	Blue
7	Brown
8	Yellow
9	Slate
10	Purple

3.7.2 *Color coding of cable WH–69B/U.*

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Cable WM-69B/U shall use the colors corresponding to the conductor sizes as tabulated below:

Conductors

AWG size (nominal)	Circular mils (nominal)	Color
24	404	White
24	404	Red
24	404	Green
24	404	Blue
18	1620	Brown
18	1620	Yellow

3.7.3 *Color coding of cable WM-70B/U.* Cable WM-70B/U shall use the colors corresponding to the conductor sizes as tabulated below:

Conductors

AWG size (nominal)	Circular mils (nominal)	Color
24	404	White
24	404	Black
24	404	Red
24	404	Green
24	404	Orange
24	404	Blue
18	1620	Brown
18	1620	Yellow

3.7.4 *Color coding of cable WM-111A/U.* Cable WM-111A/U shall use the colors corresponding to the conductor sizes as tabulated below:

Conductors

AWG size (nominal)	Circular mils (nominal)	Color
24	404	White
24	404	Black
24	404	Red
24	404	Green
24	404	Blue
18	1620	Brown
18	1630	Yellow

3.8 Staycords. Each cable shall have staycords as follows:

3.8.1 *Staycords for cables WD-27A/U and WM-69B/U.* Cable WD-27A/U and cable WM-69B/U, shall contain two staycords each of which shall consist of a ply yarn of saponified acetate rayon with a tightly applied, close braid of nylon yarn overall. Each staycord shall have a minimum breaking strength of 25 pounds.

3.8.2 *Staycord for cables WT-15A/U, WF-11A/U, WM-59A/U, WM-60A/U, WM-61A/U, WM-62A/U, WM-64A/U, WM-70B/U and WM-111A/U.* These cables shall contain one staycord which shall consist of a ply yarn or cord of saponified acetate rayon with a tightly applied, close braid or braids of nylon yarn overall. The staycord for these cables shall have a minimum breaking strength of 60 pounds.

3.9 Cabling. Cabling shall be as follows.

3.9.1 *Cable WD-27A/U.* The insulated conductors of cable WD-27A/U shall be twisted together with two staycords in the configuration indicated in figure 1. The maximum length of lay of the insulated conductors shall be one-half inch.

3.9.2 *Cables WT-15A/U, WF-11A/U, WM-59A/U, WM-60A/U, WM-61A/U, WM-62A/U, WM-63A/U, and WM-64A/U.* The insulated conductors of these cables shall be concentrically twisted around a central staycord, with the conductors in the same order around the staycord as that tabulated for the color code order specified in 3.7.1. The length of lay of the insulated conductors shall be as follows:

	Length of lay (Max. in.)
WT-15A/U	½
WF-11A/U	⅞
WM-59A/U	1⅛
WM-60A/U	1¼
WM-61A/U	1¼
WM-62A/U	1⅝
WM-63A/U	1¾
WM-64A/U	2¼

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3.9.3 *Cable WM–69B/U.* The insulated conductors of cable WM–69B/U shall be twisted together with two staycords in the configuration indicated in figure 2. The maximum length of lay of the insulated conductors shall be seven-eighths of an inch.

3.9.4 *Cable WM–70B/U.* The insulated conductors of cable WM–70B/U shall be twisted around a central staycord in the configuration indicated in figure 3. The maximum length of lay of the insulated conductors shall be seven-eighths of an inch.

3.9.5 *Cable WM–111A/U.* The insulated conductors of Cable WM–111A/U shall be twisted with fillers around a central staycord in the configuration indicated in figure 4. The fillers shall be a ply yard or braid of acetate rayon, saponified acetate rayon, or nylon. The maximum length of lay of the insulated conductors shall be seven-eighths of an inch.

3.10 Outer separator. A separator shall be applied over the cabled conductors. The separator shall consist of a suitable number of ends of acetate rayon, saponified acetate rayon, or nylon yarn closely wound to cover the cabled conductors. The yarn shall be loosely applied over the cable conductors.

3.11 Jacket. A smooth, dense, Buna S jacket, Type JS, conforming to Specification MIL–I–3930 shall be applied over the outer separator. The minimum jacket thickness shall be 0.015 inch for cable WD–27A/U and 0.020 inch for the remaining cables. The outside diameters of the cables shall be as tabulated in 3.2 with a tolerance of plus 0.010 inch and minus 0.010 inch.

3.12 Cable tensile characteristics. Cable WD–27A/U shall have a breaking strength of not less than 50 pounds. The remaining cables shall have a breaking strength of not less than 60 pounds. Continuity of all conductors shall be maintained up to at least the minimum breaking strength specified herein.

3.13 Identification marking. Cables shall be identified by ink markings. Inked markings shall be clear and legible and be resistant to moisture and oil and shall not readily abrade when the cable is stepped or tramped upon or dragged on the ground as may be required in service. The marking shall show no indication of cracking when subjected to the cold bend test. The inked marking shall consist of the type designation followed by the number of this specification and year of manufacture. Serial number indicating footage may be used at the option of the manufacturer. The marking shall be repeated along the cable at intervals not more than one foot apart.

3.14 Electrical requirements.

3.14.1 *D.c. resistance.* Each conductor in the finished cable shal be continuous and shall have a d.c. resistance per 1,000 feet at or corrected to 20°C. as follows:

Conductors		D-C resistance
AWG size (nominal)	Circular mils (nominal)	Ohms, max/1,000 ft.
24	404	39.0
18	1620	9.7

If the resistance value is lower or equal to that specified when the measurement is made at a temperature greater than 20°C., no correction factor need be employed.

3.14.2 *Dielectric strength.* The insulated conductors shall withstand without breakdown an applied voltage of 500 volts (r.m.s.) for the immersed test and 1,000 volts (r.m.s.) for the dry test.

3.14.3 *Insulation resistance.* Immediately after the insulated conductors have withstood the dielectric strength test, the insulation resistance shall be measured as specified in 4.13. The insulation resistance of the insulated conductors shall be not less than 00 megohr — 1,000 ft. at or corrected to

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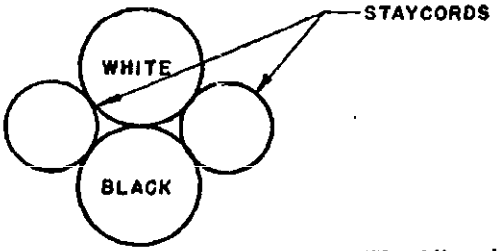


FIGURE 1. *Special purpose cable WD–27A/U cabling diagram.*

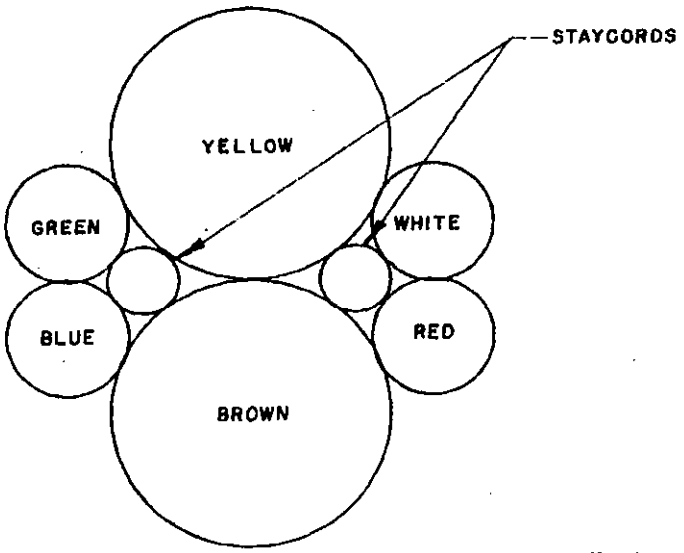


FIGURE 2. *Special purpose cable WM–69B/U cabling diagram.*

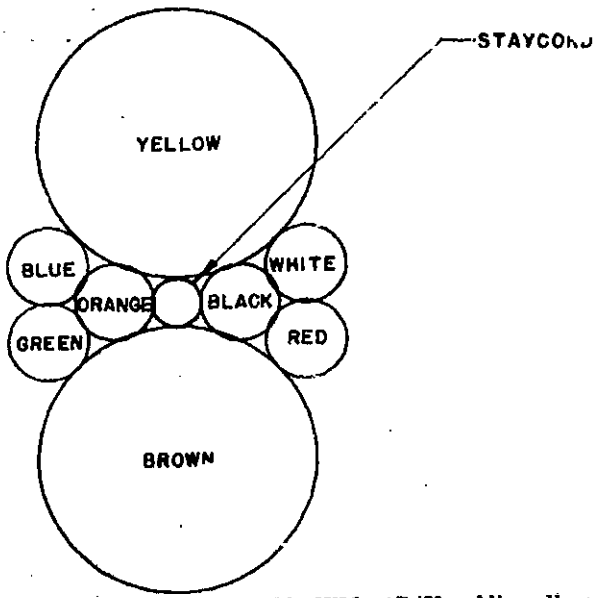


FIGURE 3. *Special purpose cable WM–70B/U cabling diagram.*

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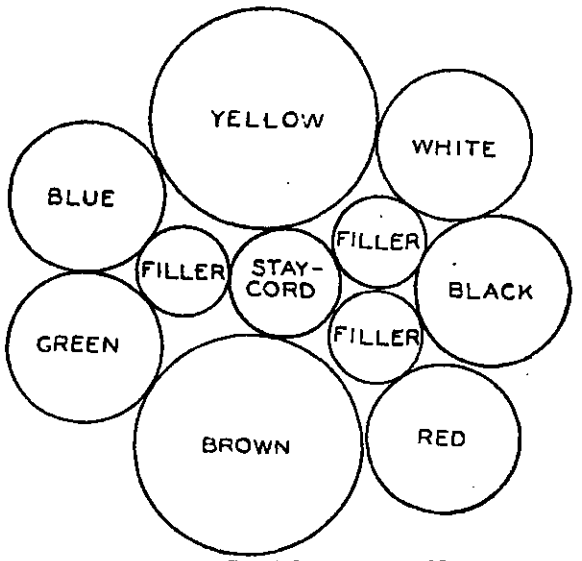


FIGURE 4. *Special purpose cable WM-111A/U cabling diagram.*

15.6°C. If the value of insulation resistance is equal to or greater than specified, when the measurement is made at a temperature greater than 15.6°C., no correction factor need be employed.

3.15 Flexing life requirements. The finished cables shall be capable of being flexed without loss of continuity of any of the conductors for not less than 30,000 cycles, when tested as specified in 4.6.

3.16 Operating range. The cables shall be flexible and resilient throughout the temperature range of —40° to +140°F.

3.16.1 *Cold bend.* Neither the jacket nor the insulation shall show evidence of cracks, flaws, or other damage when tested at —40°C. in accordance with 4.15.

3.17 Workmanship. The cables shall be manufactured in such a manner as to fulfill the requirements of 3.3 through 3.16 of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the manufacturer is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the manufacturer may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Inspection, general.

4.2.1 *Definitions.*

4.2.1.1 *Inspection lot.* An inspection lot shall be all cables of any one type produced under substantially the same conditions and offered for inspection at one time.

4.2.1.2 *Lot size.* The lot size shall be the number of units of product in the lot. A unit of product shall be defined as 1,000 feet of cable.

4.2.1.3 *Sample.* The sample shall be defined as the number of specimens to be subjected to test.

4.2.1.4 *Specimen.* The specimen shall be the individual piece of cable, selected in an unbiased manner, which is subjected to a test.

4.2.1.5 *Finished cable.* The finished cables are considered those cables on which all manufacturing operations have been completed and which are ready to be submitted for inspection.

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room ambient temperature, pressure, and humidity.

4.4 Acceptance inspection.

4.4.1 *Cable.* Acceptance inspection shall consist of groups A and B inspection.

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4.4.1.1 *Resubmitted lots.* If an inspection lot is rejected, the manufacturer may rework the lot or screen out defectives and resubmit it for the acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lot shall be inspected, using tightened inspection.

4.4.1.2 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table I. Statistical sampling and inspection shall be in accordance with MIL-STD-105.

4.4.1.2.1 When MIL-STD-105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government, or by the contractor under supervision of the Government inspector with final responsibility for acceptance of the item resting with the Government inspector.

4.4.1.3 *Group B inspection.* Group B inspection shall consist of the examinations and tests specified in table II.

4.4.1.3.1 *Sampling procedure.* The sampling procedure shall be in accordance with the Appendix to MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL shall be 6.5 percent and the inspection level shall be L-7 for normal and tightened inspection, and L-5 for reduced inspection.

4.4.2 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall conform to table III and IV.

4.4.3 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in table III and IV and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to MIL-STD-105. Inspection level L-8 shalll be used for normal inspections, and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be 4 percent and the AQL for minor defects shall be 10 percent.

4.4.3.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.4.3.2 *Procedure in case of failure.* If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken.

4.4.4 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with MIL-STD-105.

4.4.5 *Rough handling test.* When rough handling test is required by the contract, the following functional tests shall be conducted to determine freedom from malfunction caused by the rough handling:

(a) D.C. resistance.

(b) Dielectric strength.

(c) Insulation resistance.

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TABLE I. *Group A inspection.*

Inspection or test	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical				
Material	3.2 thru 3.13 incl.	4.5	1.0% for the group.	4.0% for the group.
Construction		4.5		
Workmanship	3.17	4.5		
Electrical **				
D.C. resistance	3.14.1	4.14	1.0% for the group.	*
Dielectric strength	3.14.2	4.12		
Insulation resistance	3.14.3	4.13		

* All electrical test failures are considered major.

** Electrical tests shall be conducted in the order shown.

TABLE II. *Group B inspection.*

Characteristic	Requirement paragraph	Inspection paragraph	Sampling plan
Subgroup 1			
Conductor strands			
Continuity of coating	3.4.1.1	4.7	4.4.1.3.1 for the subgroup.
Adhesion of coating	3.4.1.2	4.8	
Tensile strength and elongation	3.4.2	4.9	
Resistance	3.4.3	4.10	
Subgroup 2			
Cable			
Tensile strength	3.12	4.11	4.4.1.3.1 for the subgroup.
Flexing life	3.15	4.6	
Subgroup 3			
Insulation and jacket			
Cold bend	3.16.1	4.15	4.4.1.3.1.

TABLE III. *Classification of defects for preservation, packaging and marking.*

Major defects	Minor defects
1. Quantity per unit package not in accordance with the contract.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Item not properly cleaned prior to preservation (when applicable).	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Any item of marking information other than 7 thru 9 listed under major defects missing or illegible.
4. Use of improper or defective material.	
5. Cushioning or padding not applied or inadequate for the protection of the barrier material from projections, sharp edges or other similar features of the item.	
6. Cushioning inadequate or not applied when required for the physical and/or mechanical protection of the item.	
7. Stock number omitted, incorrect or illegible.	
8. Nomenclature incorrect or missing.	
9. Quantity of items in package not indicated.	

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TABLE IV. *Classification of defects for packing and marking.*

Major defects	Minor defects
1. Box closure not in accordance with specification. 2. Type, grade, class and style of the shipping container do not meet the minimum requirements. 3. Strapping when required is omitted or inadequate, or incorrectly applied. 4. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement and/or damage. 5. Gross weight in excess of contract limitation. 6. Use of improper or defective material. 7. Shipping documents or packing list omitted (when required). 8. Stock number (when required) omitted or illegible. 9. Nomenclature (when required) omitted or illegible. 10. Quantity incorrect or missing. 11. Destination marking incorrect or illegible. 12. Service designation (color marking) missing. 13. Special marking and labeling requirements cited in the contract, such as MAP labels, shipment digit markings, etc, not applied. 14. Overseas code marking — illegible, incomplete or missing.	1. Unsealed carton. 2. Defective taping or sealing of carbon. 3. Any other box defect which may be considered minor by definition of MIL-STD-105. 4. Any item of required marking information, other than 8 thru 14 listed under major defects, missing or illegible.

4.5 Visual and mechanical inspection. The finished cables shall be given a visual and mechanical inspection for conformance with the requirements of 3.2 to 3.13 inclusive. Visual and mechanical defects shall be classified as major or minor in accordance with the definitions of MIL-STD-105, where the Government will judge the classification for each defect found.

4.6 Flexing life test. The finished cables shall be tested for compliance with 3.15. The cable flexing machine shall contain a rotating actuating arm whose normal zero position is vertical. The arm shal rotate about a horizontal axis, the outer end traversing a circular arc in a vertical plane. The arm shall rotate 90° from the zero position to a position 90° from the zero position in the opposite direction, and then reverse its direction and rotate back to zero position, thus completing a cycle. The arm shall be rotated at approximately 900 cycles per hour. Two adjustable positions ½ inch diameter mandrels shall be mounted in the machine with their axis perpendicular to the plance of rotating of the actuating arm. The plane through the axis of the two mandrels shall be horizontal and shall pass through the axis of rotation of the actuating arm. The mandrels shall be placed so that the axis of rotation lies midway between the mandrels. A length of cable shall be inserted between the mandrels which shall be so spaced that there is a slight clearance between the cable and the mandrels. The upper end of the cable shall be attached to the rotating end of the actuating arm through a spring. The lower end of the cable shall be attached to the frame of the machine so that the cable is vertical when the actuating arm is in the zero position. The spring shall be extended when the actuating arm is in the horizontal position such that a 5 pound pull is exerted on the cable. When the arm is in the vertical position, the pull shall be not less than 4.5

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pounds. The cable shall be attached to the spring and the frame of the machine at least 5 inches from the plane through the axis of the mandrels. The cable shall have all of its conductors connected in series through a relay which will cause the machine to cease operation if any of the conductors momentarily lose continuity. A bending machine such as that described in Specification MIL-C-13777 is a suitable cable flexing machine for use as indicated above. Any other approved machine which will give equivalent mechanical motion to the cable may be used.

4.7 Continuity of coating. This test shall be made in accordance with Specification MIL-W-3861.

4.8 Adhesion of coating. This test shall be made in accordance with Specification MIL-W-3861

4.9 Conductor strand tensile strength and elongation. This test shall be made in accordance with Specification MIL-W-3861.

4.10 Conductor strand resistance. The conductor strand, prior to stranding into a conductor, shall be measured for d.c. resistance in accordance with Method 6021 of Specification JC-98.

4.11 Cable tensile strength. The tensile strength shall be made on a tensile testing machine, power driven, and preferably of the pendulum type. The test shall be made at the rate of 12 inches per minute. The conductors of the cable may be connected in series to an indicating or signaling device or automatic stop or both. The arrangement shall be such that an indication will be given when there is a disruption of electrical continuity due to the breaking of the conductors of the cable under tensile test.

4.12 Dielectric strength test. This test shall be performed in accordance with Method 6111 of Specification J-C-98 except for the following:

(a) That the voltages specified in 3.14.2 shall be applied for a 1-second duration.

(b) That the immersion be omitted if the dry test is used.

4.12.1 *Alternate test procedure.* The following alternate test procedure may be used in place of the test procedure specified above.

(a) Arrange the conductors in a rectangle containing rows and columns as specified below. The last column will contain unused spaces as specified below.

No. of conductors	*No. rows*	*No. columns*	*No. unused spaces*
3	2	2	1
4	2	2	0
5	2	3	1
6	2	3	0
7	3	3	2
8	3	3	1
9	3	3	0
10	3	4	2

(b) Connect all the conductors in a column together. Do this for each column.

(c) Apply the specified test voltage, for the specified time, between each column of the rectangle in turn, and the remaining columns connected together and also connected to the water for the immersed test.

(d) Connect all the conductors in a row together. Do this for each row.

(e) Apply the specified test voltage, for the specified time, between each row of the rectangle in turn, and the remaining rows connected together.

(f) Connect all the conductors together.

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4.13 Insulation resistance test. This test shall be performed in accordance with Method 6031 of Specification J–C–98.

4.14 D.C. resistance. This test shall be performed in accordance with Method 6021 of Specification J–C–98.

4.15 Cold bend.

4.15.1 *Specimens.* One specimen from each sample unit shall be prepared for test. Each specimen shall be divided into two parts, one for checking cable as a whole and the other for checking insulation apart from the cable.

4.15.2 *Procedure.* The specimens selected for checking cable as a whole shall be attached to the proper size mandrel as specified in 4.15.2.1. The specimens shall be suspended vertically with lower ends weighted sufficiently to keep specimens taut to and to permit bending them without handling. The mandrel and specimens shall be placed for at least 20 hours in the cold chamber at the specified temperature (see 3.16.1). While at this temperature, the specimens shall be bent for seven close turns around the mandrel at the rate of approximately fifteen turns per minute. After the test has been completed, the jacket on the specimen of cable shall be examined for damage through a magnifying glass of at least three diameter's magnification. The jacket shall be carefully removed and the individual conductor insulation of all specimens shall be examined for damage with the magnifying glass. Only the five inside turns shall be considered for evidence of damage.

4.15.2.1 *Mandrels for cables.* The mandrel for testing the cable shall be selected from the following standard size mandrel diameters.

Mandrel diameters (inches)
.500
.68
.84
1.05

The size selected shall be the largest size which does not exceed three times the specified nominal diameter of the cable.

4.15.2.2 *Mandrels for insulation.* The mandrel for testing the insulation apart from the cable shall be selected according to the size of the conductor as follows:

Conductor size (AWG)	*Mandrel diameter (inches)*
24	0.062
18	.094

4.16 Insulation and jacket. The insulation and jacket shall be examined and tested for conformance with Specification MIL–I–3930.

5. PREPARING FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Cables, special purposes, electrical shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and further described in the following paragraphs.

5.1.1.1 *Cleaning.* Dirt and other foreign material shall be removed from the finished cable by any cleaning process that will not damage the cable.

5.1.1.2 *Preservative.* Preservative shall not be applied to the cable.

5.1.1.3 *Packaging.* The finished cable shall be supplied in lengths of 500 or 1,000 feet, as specified by the procuring activity. Cable shall be uniformly wound on suitable type, size and kind of spools, in accordance with the supplier's commercial practice in such a manner, to prevent injury to the cable, and to afford maximum protection against damage. Both ends of the cable shall be securely fastened to the flanges of the spool in a manner that assures accessibility for testing purposes. The exposed ends of the cable shall be sealed with rubber tape applied over

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and back from the end approximately 1½ inches.

5.1.1.4 *Cable on spool.* Each spool of cable shall be packaged Method IC–5, of Specification MIL–C–116. Place each spool of cable in a snug fitting, water-resistant, fiberboard box, Style RSC, Class 2, compliance symbol V3c, W5c, or W6c, as applicable conforming to the requirements of Specification PPP–B–636. Fill all voids with cells or pads or both, fabricated of double faced, corrugated fiberboard, minimum dry bursting strength of 175 pounds, conforming to the water-resistant requirements of the box specification. Placement of the fiberboard shall be accomplished in such a manner as to prevent movement of the spool within the box after closure. Fiberboard box with contents, having a gross weight of 40 pounds or less, shall be sealed at all seams and joints with a 2-inch width of water-resistant, pressure sensitive tape, conforming to Specification PPP–T–76, extended over all corners and edges at least 2½ inches. Fiberboard box with contents, having a gross weight of over 40 pounds, shall be sealed in accordance with the appendix of Specification PPP–B–636, and in addition, all seams and joints shall be covered with 2-inch width of water-resistant, pressure sensitive tape, conforming to Specification PPP–T–76, extended overall edges at least 2½ inches.

5.1.2 *Level B (see section 6).* Cables, special purpose, electrical shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and further described in the following paragraphs.

5.1.2.1 *Packaging.* Lengths of cable shall be packaged following the procedure specified in 5.1.1.3.

5.1.2.2 *Cable on spool.* Each spool of cable shall be packaged Method III, of Specification MIL–C–116. Overwrap the cable with flexible, corrugated, fiberboard, Type I, conforming to Specification LLL–F–291, cut to snugly fit the distance between the spool flanges. The wrap shall overlap on itself, a minimum distance of 4 inches. With the wrap tightly drawn around the cable, secure it adequately with 1½-inch width strip(s) of pressure sensitive tape, conforming to Specification PPP–T–76.

5.1.3 *Level C.* Cables, special purposes, electrical shall be packaged in accordance with suppliers' commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage, during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* Spools of cable, packaged as specified in 5.1.1, shall be packaged in a wirebound box, Style 3, Class 3, Type II load, conforming to Specification PPP–B–585, when the gross weight of the total quantity packaged is 40 pounds or greater. Packaged quantities less than 40 pounds shall be packed in a fiberboard box, Style RSC, Type I, Class 2, compliance symbol V3c, conforming to Specification PPP–B–636. Box sizes shall be compatible with the quantity of packages contained therein. Packages shall be arranged within the box to provide maximum compactness and good balance. Void spaces shall be compensated by blocking with cells or pads or other media, fabricated from water-resistant fiberboard, compliance symbol W6c, conforming to Specification PPP–B–636, to prevent shifting within the box. The gross weight of the wirebound box shall not exceed 200 pounds. Box closure shall be in accordance with the appendix of the applicable box specification. The following alternate shipping containers may be used: PPP–B–591 (overseas type), PPP–B–601 (overseas type), and PPP–B–621 (class 2).

5.2.1.1 *Metal strapping and tape banding.* Strapping and banding shall be applied only for direct shipments to Army terminal cen-

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ters or aerial Ports of Embarkation, as described in the following paragraphs.

5.2.1.1.1 *Wirebound boxes.* Flat steel strapping, conforming to Specification QQ–S–781, Type I, Class b, shall be applied in accordance with the appendix of Specification PPP–B–585.

5.2.1.1.2 *Fiberboard boxes.* Filament tape bands, conforming to Specification PPP–T–97, Type III, ½-inch width, shall be applied with sufficient hand tension to assure adherence to the entire box surfaces. Band(s) shall circumvent the box, and overlap beyond the initial starting end, a minimum of 4 inches. The number and location of the band(s) shall be the same as required for metal strapping in accordance with the appendix of Specification PPP–B–636.

5.2.2 *Level B.* Spools of cable, packaged as specified in 5.1.2, shall be packed in a wirebound box, style 2, Class I, Type II load, conforming to Specification PPP–B–585, when the gross weight of the total quantity packaged is 65 pounds or greater. Packaged quantities less than 65 pounds shall be packed in a fiberboard box, Style RSC, Type I, Class 2, conforming to Specification PPP–B–636 as applicable. Box sizes shall be compatible with the quantity of packages contained therein. Packages shall be arranged within the box to provide maximum compactness and good balance. Void spaces shall be compensated by blocking with cells or pads or other media, fabricated from double faced, corrugated fiberboard, having a minimum dry bursting strength of 125 pounds, to prevent shifting within the box. The gross weight of the wirebound box shall not exceed 200 pounds. Box closure shall be in accordance with the applicable box specification or appendix thereto. The following alternate shipping containers may be used: PPP–B–591 (domestic type), PPP–B–601 (domestic type), PPP–B–621 (class 1), and PPP–B–640.

5.2.3 *Level C.* Spools of cable, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or Motor Freight Regulations whichever is applicable.

5.3 Marking.

5.3.1 *Normal marking.* Unit packages and shipping containers shall be marked in accordance with the requirements of MIL–STD–129.

5.3.2 *Special marking.* As specified by the contract or order.

6. NOTES

6.1 Intended use. The cables covered by this specification are intended for use with microphones, receivers, headsets, handsets, etc., when small diameter and flexibility are advantageous.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 3.2).

(c) Length of finished miniature cables to be delivered (see 5.1.1.3).

(d) Level of packaging and level of packing required for shipments (level A, level B, or level C).

(e) The specific paragraphs of section 5 which are applicable to the particular procurement.

(f) Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined to be appropriate. Effective period of protec-

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tion is 12 months from date of initial packaging.

6.3 **Storage surveillance.** Items preserved and packaged in accordance with level B requirements must be inspected to determine condition when not used within the time period indicated. Items not used within the time period specified must either be reserved or repackaged in accordance with level B requirements in this specification or with level A requirements if storage beyond an additional year is anticipated.

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